Assessing the Artificial Intelligence Landscape in G7 Nations: A CRITIC-TOPSIS Multi-Criteria Decision-Making Approach

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Abstract: The proliferation of Artificial Intelligence (AI) presents a dual-edged challenge for developed nations, creating immense economic opportunities alongside complex societal risks. This paper addresses the critical need for a holistic assessment framework by proposing and applying an integrated Multi-Criteria Decision-Making (MCDM) model to the Group of Seven (G7) ecosystems. We employ the objective CRITIC method to determine the weights of 13 quantitative indicators across four pillars, which reveals that data privacy regulations and social inequality are the most statistically significant differentiators among the G7. Subsequently, the TOPSIS method ranks the nations' overall performance. Findings reveal the United States ranks first, propelled by dominant Economic and Tech Readiness scores, while the United Kingdom emerges in second, showcasing a more balanced profile with a leading performance in Social Impact. A scenario analysis confirms the robustness of the top-tier rankings, highlighting their stability across divergent policy priorities. This study contributes a novel and replicable MCDM framework for benchmarking national AI capabilities, offering data-driven insights for policymakers to navigate the strategic trade-offs of the AI era.

Keywords: Artificial Intelligence (AI), Multi-Criteria Decision-Making (MCDM), CRITIC-TOPSIS, G7 Countries, Technology Assessment, National AI Strategy.

1. Introduction

1.1. Background

The 21st century has witnessed the rapid development and widespread adoption of Artificial Intelligence (AI), a general-purpose technology that is profoundly transforming industries, economies, and social structures worldwide. AI is no longer a futuristic concept but has become a foundational force driving breakthroughs in scientific discovery, supply-chain optimization, personalized healthcare, and national security [1], [2]. According to PwC, AI could contribute up to USD 15.7 trillion to the global economy by 2030, highlighting its transformative potential across all sectors [1].

Amid this technological revolution, the Group of Seven (G7) nations - Canada, France, Germany, Italy, Japan, the United Kingdom, and the United States - stand at the forefront of AI development. Collectively, G7 countries account for more than 60% of global private AI investment and host several of the world’s leading research institutions [2]. Each G7 member recognizes the strategic importance of AI and has formulated distinct national strategies to foster innovation while addressing risks related to ethics, employment, and social inequality.

The interplay of competition and cooperation among these nations creates a complex geopolitical landscape. AI has emerged as a new domain of strategic competition, particularly between democratic and authoritarian systems. Consequently, developing a clear and comprehensive understanding of each country’s position - not only in terms of economic investment, but also across the full spectrum of technological readiness, societal impact, and governance capacity - has become essential.

1.2. Research gap and motivation

Although numerous reports and indices monitor isolated indicators of AI - such as total private investment, the number of patents filed, or academic publication counts - there remains a significant research gap in providing a holistic and integrated assessment of national AI ecosystems. Existing comparative studies are often qualitative, rely on expert judgments, or focus only on a few dimensions such as technological strength or economic capacity.

Such a fragmented approach risks leading to incomplete or misleading conclusions, overlooking the complex interplay among factors such as public policy, social adoption, and technical infrastructure [3], [4].

For example, a country may lead in the number of research publications yet lag in technology commercialization or establishing a legal framework protecting citizen privacy. A single - indicator approach will fail to capture this broader picture. Hence, there is an urgent need for a quantitative, multi-dimensional framework capable of objectively assessing and comparing the overall performance of these complex ecosystems. The motivation for this research springs from the desire to fill that void, in order to provide policymakers with a sharper and more comprehensive analytical tool to make strategic, data-driven decisions in the context of the growing AI era.

1.3. Novelty and contribution

This study makes several novel and significant contributions to both academia and policy formulation.

First, we propose a comprehensive and structured AI ecosystem assessment framework, consisting of 13 carefully selected quantitative indicators grouped into four key pillars: Economic Impact, Social Impact, Policy & Governance, and Technological Readiness. This framework goes beyond traditional measures-such as R&D investment or publication counts - to provide a holistic 360 - degree view of a nation's AI capability [1].

Second, we adopt an integrated CRITIC - TOPSIS multi-criteria decision-making (MCDM) methodology. The main innovation lies in the application of the CRITIC (Criteria Importance Through Intercriteria Correlation) method to derive objective indicator weights directly from the internal correlations of the data. Unlike expert-opinion-based weighting, this approach reduces subjectivity and enhances the objectivity and robustness of the assessment [5].

Third, this research represents one of the first applications of an MCDM approach to evaluate and rank the AI ecosystems of the G7 economies comprehensively. By focusing on these technologically advanced nations, the study provides a valuable benchmarking tool, enabling direct comparison and the identification of best practices in AI governance and innovation [6].

Finally, through a detailed case study and scenario analysis, we deliver actionable insights for both academia and policymakers. The findings not only present the final ranking results but also uncover each G7 country's strengths and weaknesses across the four pillars-offering strategic guidance for nations to improve their AI readiness and competitiveness in the global landscape.

2. Materials and Methods

This chapter outlines the comprehensive analytical framework developed to assess and rank the artificial intelligence (AI) ecosystems of the G7 nations. It details the research process, the criteria selected for evaluation, the data sources utilized, and the theoretical underpinnings of the multi-criteria decision-making (MCDM) methods employed: the CRITIC method for objective weighting and the TOPSIS method for performance ranking.

2.1. Analytical Framework

The evaluation of national AI ecosystems presents a classic multi-criteria decision-making (MCDM) problem. It requires the assessment of multiple alternatives (the G7 nations) against numerous, often conflicting, quantitative criteria, ranging from economic investment to social equity. A simplistic approach that relies on single metrics or subjective weighting is inadequate to capture the complex interplay and trade-offs inherent in this domain. Such a structured approach is essential for synthesizing diverse indicators into a coherent and defensible overall assessment. The analytical framework for this study is therefore built upon an integrated, two-phase MCDM model, specifically designed to ensure objectivity and provide a holistic perspective. This framework moves beyond merely ranking the countries; it first seeks to understand the intrinsic importance of the factors that differentiate them.

The first phase of the framework is dedicated to Objective Weighting. The core principle here is that before performance can be ranked, the relative importance of each evaluation criterion must be determined. To eliminate the subjective biases often associated with expert-based weighting panels (e.g., AHP), this study employs the CRITIC (CRiteria Importance Through Intercriteria Correlation) method. The selection of CRITIC is deliberate: it is a data-driven technique that derives the weight of each criterion directly from the statistical properties of the data itself. By quantifying both the contrast intensity (standard deviation) and the degree of conflict (correlation) of each criterion, CRITIC provides an objective measure of the information content it contributes to the decision-making problem.

The second phase of the framework is Performance Ranking. Once the objective weights are established, the TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution) method is applied to evaluate and rank the G7 nations. TOPSIS is a robust and widely-used MCDM technique that ranks alternatives based on their simultaneous shortest distance from an ideal solution and farthest distance from a negative-ideal solution. This sequential integration of CRITIC and TOPSIS creates a powerful synergy: CRITIC first provides an objective lens on *what matters most* within the G7 context, and TOPSIS then uses that lens to measure *how well each nation performs*. Ultimately, this CRITIC-TOPSIS framework provides a holistic and transparent analytical tool that not only delivers a final ranking but also reveals the underlying structure and key drivers of the G7's AI landscape, directly addressing the research gap for a more nuanced and evidence-based assessment model.

2.2. Framework and Criteria Selection

The credibility of this assessment framework rests upon the careful and systematic selection of its core components, which was guided by an extensive review of established literature. The framework is built upon a two-level hierarchy: four high-level pillars that define the essential dimensions of a national AI ecosystem, and 13 specific quantitative criteria nested within them. This structure is designed to be both comprehensive in scope and granular in detail. Table 1 summarizes the 13 selected criteria along with their respective data sources. The raw data obtained from these sources were processed and integrated to form the final dataset used in this analysis, as shown in Table 14.

Table 1. The Assessment Framework Criteria

Label	Criteria Name	Pillar	Type
C1	Private Investment in AI (billions of U.S. dollars) [21]	Economic Impact	Benefit
C2	Number of AI Startups per 1 million inhabitants [22][24]	Economic Impact	Benefit
C3	Average Labor Productivity Growth (%) [15]	Economic Impact	Benefit
C4	Share of Jobs at High Risk of Automation (%) [14]	Economic Impact	Non-benefit
C5	Share of high digital-intensive sectors in total employment (%) [13]	Social Impact	Benefit
C6	Income inequality: Gini coefficient [16]	Social Impact	Non-benefit
C7	AI Adoption in Public Healthcare [23]	Social Impact	Benefit
C8	Maturity of the National AI Strategy [7]	Policy & Governance	Benefit
C9	Data protection privacy law index [25]	Policy & Governance	Benefit
C10	Government AI readiness index score [17]	Policy & Governance	Benefit
C11	Fixed Broadband Speed (Mbps) [12]	Tech Readiness	Benefit
C12	AI publications by country (%) [11]	Tech Readiness	Benefit
C13	Percentage of SMEs using/have applied AI (%) [10] [19] [9] [8][20] [18]	Tech Readiness	Benefit

For clarity and conciseness in the subsequent analysis and tables, these criteria will henceforth be referred to by their designated labels (e.g., C1 for Private Investment in AI, C2 for Number of AI Startups, and so on).

The rationale for selecting the four core pillars-Economic Impact, Social Impact, Policy & Governance, and Tech Readiness-is based on their prevalence and comprehensiveness in established global indices and academic research. As demonstrated by the literature survey presented in Table 2, these four pillars consistently appear as the foundational components required for a holistic national assessment. This approach deliberately opts for broader, more encompassing dimensions over more specialized ones. While granular pillars such as "AI Ethics & Safety" or "National Security" are critically important in specific contexts, our review of the literature indicates that their core elements are often integrated within, or considered outcomes of, the broader governance and social dimensions. For instance, ethical considerations are a fundamental component of a mature governance framework, and a nation's security posture is deeply influenced by its technological readiness and economic strength. By adopting these four pillars, the study creates a robust structure that captures the multifaceted nature of an AI ecosystem in a balanced and comprehensive manner, avoiding both redundancy and oversimplification.

Table 2. Selection of Assessment Pillars Based on Prior Studies

No.	Authors/Organization	Year	Economic Impact	Social Impact	Policy & Governance	Tech Readiness
1	Stanford University [2]	2025	X	X	X	X
2	Tortoise Media [27]	2024	X	X	X	X
3	Castro, D., & McLaughlin, M. [28]	2021	X	X		X
4	Oxford Insights [29]	2023		X	X	X
5	OECD.AI [30]	2024	X	X	X	
6	NSCAI [31]	2021		X	X	X
7	UNESCO [32]	2021		X	X	X
8	West, D. M. & Allen, J. R. [33]	2020	X	X	X	X

Following the establishment of the four pillars, specific quantitative criteria for each were selected based on their frequent use in key literature, ensuring that the chosen indicators are both relevant and widely accepted as valid measures within their respective domains.

2.2.1. The Economic Impact Pillar

This pillar is designed to measure the extent to which AI is being integrated into a nation's economy, driving growth, and fostering a vibrant commercial environment. As detailed in the literature review in Table 3, the four chosen criteria provide a balanced view of both the opportunities and the disruptive forces of AI. Private Investment in AI (C1) is widely considered the primary signal of commercial vitality, investor confidence, and the overall health of the private-sector AI ecosystem. The Number of AI Startups (C2) serves as a crucial proxy for entrepreneurial dynamism, indicating a nation's capacity to generate new, innovative firms that can challenge incumbents and create new markets. To measure the tangible macroeconomic benefits, Average Labor Productivity Growth (C3) is a standard indicator used to assess the real-world impact of AI adoption on economic efficiency. Finally, to capture the potential structural challenges and social costs, the Share of Jobs at High Risk of Automation (C4) provides a critical forward-looking metric that is essential for understanding the transformative and often disruptive nature of AI on the labor market. The selection of these four indicators is thus strongly supported by their frequent and combined use in key economic and technology reports.

Table 3. Use of Economic Impact Criteria in Key Literature

No.	Authors	Year	Criteria			
			C1	C2	C3	C4
1	Stanford University (AI Index) [2]	2025	X	X		
2	Castro, D., & McLaughlin, M. (AI Race) [28]	2021	X	X		
3	Tortoise Media (Global AI Index) [27]	2024	X	X		
4	PwC (Sizing the Prize) [34]	2018	X		X	
5	OECD (AI in Society) [35]	2019			X	X
6	World Economic Forum (Future of Jobs) [36]	2023			X	X
7	Frey, C. B., & Osborne, M. A. [37]	2017				X
8	Acemoglu, D., & Restrepo, P. [38]	2019			X	X
9	Roland Berger [39]	2022		X		

2.2.2. The Social Impact Pillar

This pillar evaluates how AI is affecting the fabric of a nation's society, including its workforce, equity, and public welfare. The three criteria chosen for this pillar, as justified by the literature review in Table 4, are designed to capture these diverse societal dimensions. The Share of high digital-intensive sectors in total employment (C5) is a key metric for gauging the structural transformation of the labor market toward a more skilled, AI-ready workforce. It reflects the extent to which the economy is shifting towards jobs that complement, rather than compete with, AI technologies. Income inequality (C6), measured by the Gini coefficient, directly addresses a primary ethical and social stability concern raised in major reports from international bodies like the OECD and UNESCO. This criterion is crucial for assessing whether the economic gains from AI are being broadly shared or are instead exacerbating existing social disparities. Furthermore, AI Adoption in Public Healthcare (C7) serves as a tangible and direct measure of how AI is being leveraged for broad societal good, moving beyond purely economic metrics to capture improvements in public well-being. Together, these criteria provide a robust framework for assessing whether AI development is translating into positive and equitable social outcomes.

Table 4. Use of Social Impact Criteria in Key Literature

No.	Authors	Year	Criteria		
			C5	C6	C7
1	OECD (Employment Outlook) [40]	2023	X	X	
2	West, D. M. & Allen, J. R. (Brookings) [33]	2020		X	
3	UNESCO (Ethics of AI) [32]	2021		X	
4	World Health Organization (WHO) [41]	2021			X
5	Vinuesa, R. et al. (AI for SDGs) [42]	2020			X
6	World Bank (World Development Report) [43]	2021			X
7	Acemoglu, D., & Restrepo, P. [38]	2019	X	X	
8	Topol, E. (Nature Medicine) [44]	2019	X	X	
9	World Economic Forum & BCG [45]	2020			X
10	Frank, M. R. et al. (PNAS) [46]	2019	X		

2.2.3. *The Policy & Governance Pillar*

This pillar assesses the crucial role of the state in steering the AI ecosystem through strategic direction, regulation, and public sector capacity. The literature review presented in Table 5 confirms that the three selected indicators are foundational for this assessment. The Maturity of the National AI Strategy (C8) is a universally used indicator to evaluate government foresight, long-term commitment, and the coherence of its approach to fostering AI. The Data protection privacy law index (C9) is critical for building public trust and serves as a proxy for a nation's commitment to ethical AI and the protection of human rights in the digital age. It reflects the maturity of the legal environment in which AI operates. Lastly, the Government AI readiness index score (C10), pioneered by institutions like Oxford Insights, directly measures the capacity of the public sector to not only regulate but also to adopt and implement AI effectively for public services, signaling a government's ability to lead by example. These indicators collectively provide a comprehensive snapshot of a nation's governance landscape for AI.

Table 5. Use of Policy & Governance Criteria in Key Literature

No.	Authors	Year	Criteria		
			C8	C9	C10
1	Oxford Insights [29]	2023	X	X	X
2	OECD.AI [30]	2024	X	X	
3	West, D. M. & Allen, J. R. (Brookings) [33]	2020	X	X	
4	Tortoise Media [27]	2024	X		
5	World Bank (GovTech) [47]	2022	X	X	X
6	Deloitte (AI in Government) [48]	2023	X		X
7	World Economic Forum (WEF) [49]	2023	X	X	X
8	UNESCO (Ethics of AI) [32]	2021	X	X	
9	van der Lely, M. et al. [50]	2021	X		
10	CSET, Georgetown University [51]	2022	X		

2.2.4. *The Tech Readiness Pillar*

This pillar measures the foundational technological infrastructure and research capacity that enable all other aspects of AI development and deployment. As demonstrated in Table 6, the three chosen criteria capture the most essential components of this critical foundation. Fixed Broadband Speed (C11) is a fundamental indicator of the digital infrastructure required for a data-intensive AI economy, as high-speed connectivity is a prerequisite for accessing and utilizing large datasets and cloud computing services. The volume of AI publications by country (C12) is the primary metric used globally to assess a nation's contribution to cutting-edge research and scientific leadership, reflecting its capacity to generate new knowledge and push the frontiers of AI. Finally, the Percentage of SMEs using AI (C13) is a crucial indicator for determining whether AI adoption is widespread and democratized across the economy or confined to a few large, leading firms. This metric measures the true breadth of technological diffusion and the overall absorptive capacity of the economy.

Table 6. Use of Tech Readiness Criteria in Key Literature

No.	Authors	Year	Criteria		
			C11	C12	C13
1	Stanford University (AI Index) [2]	2024		X	
2	OECD (Digital Transformation of SMEs) [52]	2021			X
3	International Telecommunication Union (ITU) [53]	2023	X		
4	Castro, D., & McLaughlin, M. (AI Race) [28]	2021		X	
5	World Economic Forum (Network Readiness) [36]	2023	X		
6	Scimago Lab (SJR) [54]	2023		X	
7	Dutta, S., Lanvin, B., & Soumitra, R. (Global Innovation Index) [55]	2023		X	
8	European Commission (Eurostat) [56]	2022			X
9	UNCTAD (Digital Economy Report) [57]	2022	X		
10	Bloom, N. et al. (NBER) [58]	2021			X

2.3. Methodology Selection: An Objective CRITIC-TOPSIS Framework

The evaluation of national AI ecosystems is a classic multi-criteria decision-making (MCDM) problem, characterized by the need to assess multiple alternatives (the G7 nations) against numerous, often conflicting, quantitative criteria. To address this complexity, a two-phase MCDM framework integrating the CRITIC and TOPSIS methods was deliberately chosen. This combination creates a powerful synergy that ensures both objectivity in determining criteria importance and robustness in the final performance ranking, and its suitability is underscored by its successful application in analogous assessment problems.

The first critical challenge in any MCDM analysis is the assignment of criteria weights. To circumvent the limitations of subjective methods, this study employs the CRITIC (CRiteria Importance Through Intercriteria Correlation) method. The selection of CRITIC is justified by its core principle of objectivity. It is a data-driven technique that derives weights based on the intrinsic information contained within the dataset itself. This is particularly crucial for a topic like AI, where preconceived notions about the importance of certain criteria (e.g., investment) might overshadow other statistically significant factors. By evaluating both the contrast intensity and the degree of conflict of each criterion, CRITIC provides an objective measure of the information it contributes to the decision-making problem.

Once the objective weights are established, the TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution) method is used for the performance ranking. TOPSIS was selected for its logical simplicity, its ability to handle a significant number of alternatives and criteria, and its computational efficiency. The fundamental logic of TOPSIS is highly intuitive and well-suited for national benchmarking: the most desirable alternative is the one that is simultaneously closest to the Positive-Ideal Solution (the best possible performance across all criteria) and farthest from the Negative-Ideal Solution (the worst possible performance).

The integration of CRITIC and TOPSIS creates a seamless and powerful analytical framework that has become a well-established and validated approach for complex national-level assessments. CRITIC first provides an objective answer to the question of "what matters most?" by identifying the key differentiators within the data. TOPSIS then takes these objective weights to provide a robust answer to the question of "who performs best?". As shown in Table 7, this hybrid model has been successfully applied across a diverse range of problems that are structurally similar to the assessment of AI ecosystems, such as evaluating national innovation capabilities, digital economy readiness, and the performance of major economies in sustainability. The proven effectiveness of this methodology in these highly relevant and analogous domains provides a solid justification for its application in the present study.

Table 7. Application of the CRITIC-TOPSIS Method in Relevant National and Technological Assessment Studies

No.	Authors	Year	Application Domain	Methodology Used
1	Li, H. et al. [59]	2022	National Innovation Capability of OECD Countries	Integrated CRITIC-TOPSIS
2	Li, H. et al. [60]	2023	Digital Economy Development Level in the EU	CRITIC-TOPSIS Approach
3	Gati, A. et al. [61]	2023	Sustainability Performance of G20 Countries	CRITIC-TOPSIS Approach
4	Stanković, J. J. et al. [62]	2021	Ranking Countries by Sustainable Development	CRITIC-TOPSIS Approach
5	Ulutaş, A. et al. [63]	2022	Measuring the Information Society Index in Europe	Integrated CRITIC-TOPSIS
6	Ecer, F. & Ayag, Z. [64]	2021	Evaluating E-Government Readiness of EU Countries	CRITIC-based TOPSIS
7	Tusnio, N. [65]	2023	Assessing Digital Transformation Readiness in ASEAN	CRITIC-TOPSIS Method
8	Wu, H. et al. [66]	2020	Evaluating R&D Performance in High-Tech Sectors	Integrated CRITIC-TOPSIS
9	Adali, E. A. & Isik, A. T. [67]	2017	Financial Performance of National Banks	Hybrid CRITIC-TOPSIS
10	Deveci, M. et al. [68]	2022	Smart City Technology Assessment	CRITIC-TOPSIS Framework

2.4. The CRITIC Method

The CRITIC method, developed by Diakoulaki, Mavrotas, and Papayannakis (1995), is an objective weighting technique that determines the relative importance of criteria based on the statistical information inherent in the data. Its core logic is that a criterion is more important if it contains more information, which is quantified through two key dimensions: the contrast intensity and the degree of conflict. The contrast intensity is measured by the standard deviation of the data for a criterion, while the degree of conflict is evaluated based on the correlation between that criterion and all others.

Let the decision matrix be denoted by X, with m alternatives (G7 nations) and n criteria (indicators):

$$X = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{12} & x_{23} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{pmatrix} \quad (1)$$

The process begins with the initial decision matrix, which comprises the raw data collected for the 7 G7 nations across the 13 selected criteria. This matrix, denoted as Equations (1), is presented in Table 14.

Step 1: Data Normalization

To eliminate dimensional differences and ensure comparability, the raw data for each criterion is transformed into a dimensionless scale. This study uses linear min-max normalization. The formula differs for benefit criteria (where higher values are better) and cost criteria (where lower values are better).

For a benefit criterion j:

$$r_{ij} = \frac{x_{ij} - x_j^{min}}{x_j^{max} - x_j^{min}} \quad (2)$$

For a cost criterion j:

$$r_{ij} = \frac{x_j^{max} - x_{ij}}{x_j^{max} - x_j^{min}} \quad (3)$$

where r_{ij} is the normalized value of the i-th alternative for the j-th criterion, and x_j^{max} and x_j^{min} are the maximum and minimum values for criterion j, respectively. The resulting normalized matrix is shown in Table 15.

Step 2: Determination of Contrast Intensity and Conflict

The contrast intensity of a criterion is represented by its standard deviation (σ_j),

$$\sigma_j = \sqrt{\frac{1}{m}\sum_{i=1}^{m}(r_{ij} - \bar{r}_j)^2} \quad (4)$$

which measures the variability among the alternatives for that criterion. The degree of conflict is evaluated based on the correlation between that criterion and all others. To quantify this conflict, the Pearson correlation coefficient (ρ_{jk}) is computed for every pair of criteria. The resulting inter-criteria correlation matrix is presented in Table 16. A criterion that has a low correlation with others provides more unique information and thus has a higher degree of conflict.

Step 3: Calculation of Information Quantity

The total amount of information (C_j) emitted by each criterion is calculated by multiplying its contrast intensity (standard deviation) by its degree of conflict, as shown in Equation (5):

$$C_j = \sigma_j \sum_{k=1}^{n} (1 - \rho_{jk}) \tag{5}$$

A higher value of C_j is indicates that criterion j contains more information and is therefore more significant in differentiating the alternatives.

Step 4: Derivation of Objective Weights

Finally, the objective weight of each criterion (w_j) is derived by normalizing its information quantity according to Equation (6):

$$w_j = \frac{C_j}{\sum_{k=1}^{n} C_k} \tag{6}$$

The application of these steps yields the final objective weights for each criterion, as detailed in Table 8. These weights, which objectively reflect the amount of information each indicator contributes to the decision-making problem, are subsequently used as inputs for the TOPSIS method.

Table 8. Objective Criteria Weights Derived from the CRITIC Method

	Standard Deviation	Conflict (Sum(1-r))	Information (C)	CRITIC Weights
C1	0.345442963	7.861092057	2.715558936	0.065963091
C2	0.377942015	7.219784334	2.728659841	0.066281322
C3	0.285037126	8.907703052	2.539026073	0.061674967
C4	0.315518646	8.10649735	2.557751067	0.062129812
C5	0.325362028	9.085558475	2.956095732	0.071805921
C6	0.311067981	16.03939338	4.989341714	0.121195086
C7	0.343385836	6.830028735	2.345335125	0.056970059
C8	0.283223268	11.63837342	3.296258154	0.080068737
C9	0.338815464	18.21684724	6.172149544	0.149926431
C10	0.278967338	6.691913234	1.866825224	0.045346672
C11	0.377392435	9.464385068	3.571787329	0.086761561
C12	0.333514498	7.933607033	2.645972963	0.064272792
C13	0.356949665	7.796878514	2.783093171	0.067603551

2.5. The TOPSIS Method:

The TOPSIS method, introduced by Hwang and Yoon (1981), is a distance-based MCDM technique used to rank a finite set of alternatives. Its fundamental principle is that the optimal alternative should have the shortest geometric distance from the Positive-Ideal Solution (PIS) and the longest geometric distance from the Negative-Ideal Solution (NIS).

Using the objective weights (w_j) derived from the CRITIC method Table 16, the TOPSIS procedure is executed as follows:

Step 1: Vector Normalization of the Decision Matrix

The initial decision matrix X (Table 8) is first normalized using the vector normalization method to create a comparable, dimensionless matrix N.

$$n_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{m} x_{ij}^2}} \tag{7}$$

The application of Equation (7) to the initial data matrix results in the TOPSIS normalized matrix, shown in Table 17.

Step 2: Construction of the Weighted Normalized Decision Matrix

The weighted normalized decision matrix V is constructed by multiplying the normalized values (n_{ij}) by their corresponding objective weights (w_j):

$$v_{ij} = w_j \times n_{ij} \tag{8}$$

This matrix, which reflects the relative performance of each nation on each criterion after accounting for its objective importance, is presented in Table 18.

Step 3: Determination of the Ideal and Negative-Ideal Solutions

From the weighted normalized matrix V (Table 18), the Positive-Ideal Solution (A^+) and the Negative-Ideal Solution (A^-) are identified.

$$A^+ = \{v_1^+, v_2^+, \dots, v_n^+\} = \{(\max_i(v_{ij}) | j \epsilon J_{ben}), (\min_i(v_{ij}) | j \epsilon J_{cost})\} \tag{9}$$

$$A^- = \{v_1^-, v_2^-, \dots, v_n^-\} = \{(\min_i(v_{ij}) | j \epsilon J_{ben}), (\max_i(v_{ij}) | j \epsilon J_{cost})\} \tag{10}$$

where J_{ben} is the set of benefit criteria and J_{cost} is the set of cost criteria. The resulting PIS and NIS values for each criterion are summarized in Table 9.

Table 9. Ideal and Negative-Ideal Solutions (PIS & NIS) for the TOPSIS Method

	Ideal Solution (PIS)	Negative Ideal (NIS)
C1	0.065849370	0.000519164
C2	0.039978200	0.003482764
C3	0.040310091	-0.002300873
C4	0.016418879	0.029618371
C5	0.030928204	0.024406009
C6	0.040885240	0.054787592
C7	0.027931870	0.011172748
C8	0.040127364	0.013375788
C9	0.069303517	0.013860703
C10	0.019047596	0.015587382
C11	0.047407638	0.014827587
C12	0.058783608	0.007342068
C13	0.037996006	0.004388271

Step 4: Calculation of Separation Measures

The n-dimensional Euclidean distance of each alternative from the PIS (S_i^+) and the NIS (S_i^-) is calculated:

Distance from the Positive-Ideal Solution:

$$S_i^+ = \sqrt{\sum_{j=1}^{n} (v_{ij} - v_j^+)^2} \tag{11}$$

Distance from the Negative-Ideal Solution:

$$S_i^- = \sqrt{\sum_{j=1}^{n} (v_{ij} - v_j^-)^2} \tag{12}$$

Step 5: Calculation of the Relative Closeness to the Ideal Solution

The relative closeness, or performance score (C_i), of each alternative i to the ideal solution is calculated:

$$C_i = \frac{S_i^-}{S_i^+ + S_i^-} \tag{13}$$

The separation distances calculated using Equation (11) and Equation (12), along with the final closeness coefficient from Equation (13), are presented in Table 10.

Table 10. Seperation Measures and Closeness Coefficient

	Code	Distance to PIS (S+)	Distance to NIS (S-)	Closeness Coefficient (Score)
Canada	CA	0.093999589	0.0644133450	0.406616704
France	FR	0.099370744	0.0712811398	0.417699109
Germany	DE	0.094147058	0.0683924674	0.420774375
Italy	IT	0.107589694	0.0610622118	0.362060609
Japan	JP	0.098441246	0.0555145366	0.360587538
United Kingdom	GB	0.086444129	0.0711358515	0.451426958
United States	US	0.057486646	0.1139952393	0.664765491

Step 6: Ranking the Alternatives

The alternatives (G7 nations) are ranked in descending order based on their relative closeness score (C_i), as shown in the final column of Table 10. The alternative with the highest C_i value is ranked as the best-performing nation in the overall AI ecosystem assessment.

3. Result

This section presents the empirical results derived from the application of the CRITIC-TOPSIS framework. The findings are organized into four subsections: the objective criteria weights determined by the CRITIC method, the final overall ranking of the G7 nations, a granular analysis of performance across the four pillars, and a robustness check through scenario analysis.

3.1. Criteria Importance

The CRITIC method was first employed to derive the objective weights for the 13 evaluation criteria based on the data's inherent contrast and conflict. The calculated weights for all 13 criteria are detailed in Table 8 and visualized in Figure 1.

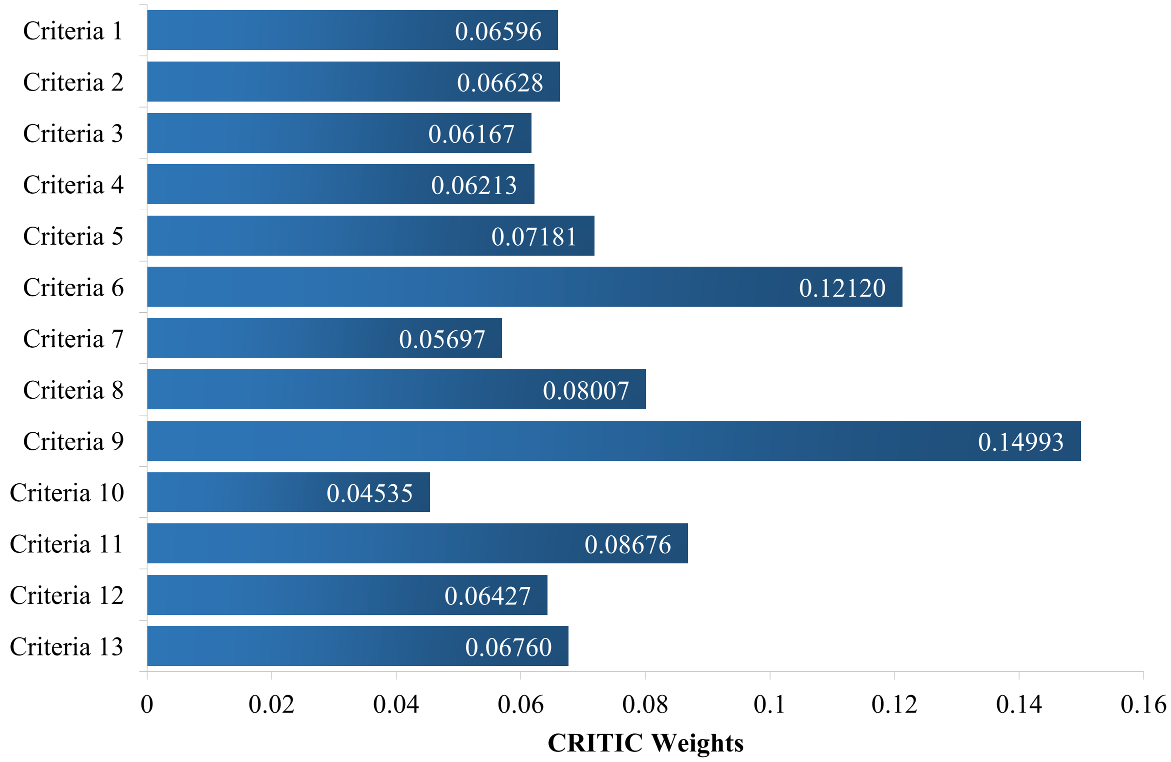

Figure 1. Objective Weights of the 13 Evaluation Criteria

A notable finding is the non-uniform distribution of importance across the criteria. Criteria 9 (Data Protection Privacy Law Index) emerged as the most influential factor with a weight of 0.1499, followed by Criteria 6 (Income inequality: Gini coefficient) with a weight of 0.1212. These two criteria, representing governance and social equity, hold significantly more weight than the others. The next most important criteria were Criteria 11 (Fixed Broadband Speed) and Criteria 8 (Maturity of the National AI Strategy), with weights of 0.0868 and 0.0801, respectively. Conversely, Criteria 10 (Government AI readiness index score) received the lowest weight (0.0454), indicating it was the least differentiating factor among the G7 countries in this dataset.

3.2. Overall Performance Ranking

Using these objective weights, the TOPSIS method was applied to the initial decision matrix to calculate the final performance score for each G7 country. The final performance scores and the corresponding overall ranks of the G7 nations are presented in Table 11 and Figure 2.

Table 11. Final Performance Scores and Overall Ranking of G7 Nations

Country	TOPSIS Score	Rank
Japan	0.36059	7
Italy	0.36206	6
Canada	0.40662	5
France	0.41770	4
Germany	0.42077	3
United Kingdom	0.45143	2
United States	0.66477	1

The United States emerges as the definitive leader, achieving a TOPSIS score of 0.6648, substantially ahead of the United Kingdom in second place (0.4514). Germany (0.4208), France (0.4177), and Canada (0.4066) form a competitive middle tier with closely grouped scores. Italy (0.3621) and Japan (0.3606) rank sixth and seventh, respectively, indicating significant room for improvement across the assessed criteria.

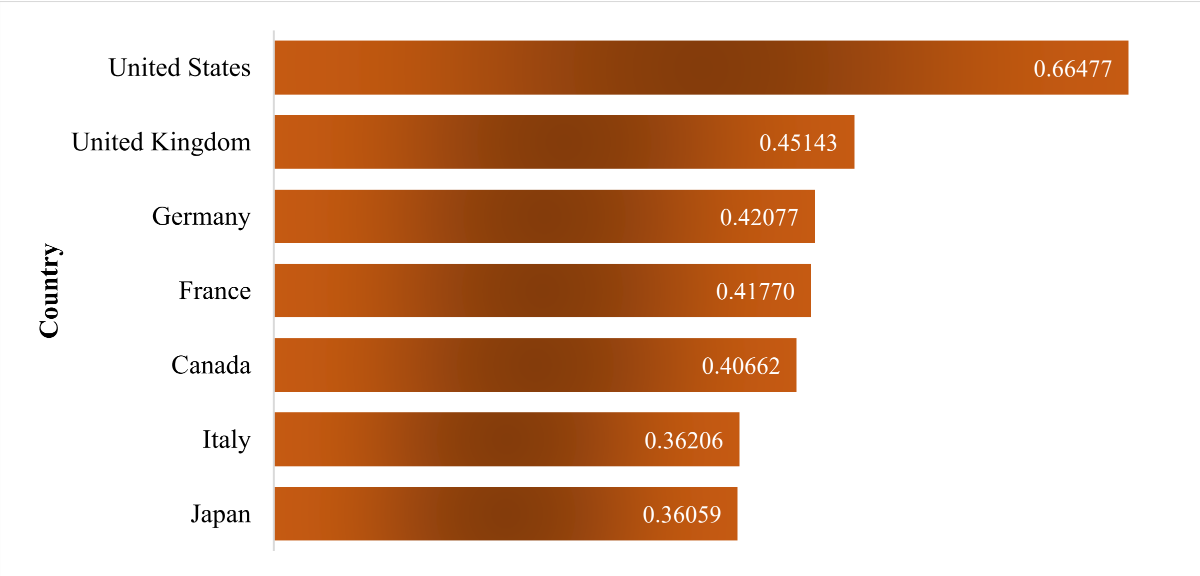

Figure 2. Overall Performance Ranking of G7 Nations based on TOPSIS Scores

3.3. Pillar-Level Analysis

To dissect the drivers behind the overall ranking and identify the specific strengths and weaknesses of each nation, a separate TOPSIS analysis was conducted for each of the four pillars. Table 12 provides a comprehensive dashboard of the country ranks across the overall score and the individual pillars.

Table 12. Pillar-Level Performance Ranking Dashboard

Country	Overall Rank	Economic Impact Rank	Social Impact Rank	Policy & Governance Rank	Tech Readiness Rank
Canada	5	3	3	6	2
France	4	7	2	2	3
Germany	3	4	5	3	6
Italy	6	6	7	1	7
Japan	7	5	6	4	5
United Kingdom	2	2	1	5	4
United States	1	1	4	7	1

The results reveal distinct national profiles. The United States' top overall rank is clearly driven by its number-one position in both Economic Impact and Tech Readiness, but this is counterbalanced by its last-place rank in Policy & Governance. Conversely, European nations lead in governance, with Italy ranking first in this pillar, but they show weaker performance in the economic and technological domains. The United Kingdom demonstrates the most balanced profile, securing the top rank in Social Impact and the second rank in Economic Impact. Canada shows a particular strength in Tech Readiness (Rank 2) but is weaker in governance.

3.4. Robustness Check via Scenario Analysis

Finally, to assess the stability of these findings under different weighting assumptions, a scenario analysis was conducted. The resulting shifts in country rankings across four scenarios-Original CRITIC, 'Economy First', 'Social First', and 'Balanced'-are summarized in Table 13 and visualized in the slope chart in Figure 3.

Table 13. Summary of Country Ranks Across Different Scenarios

Country	Economy First	Social First	Balanced	Original CRITIC
Canada	3	3	3	5
France	6	4	4	4
Germany	4	5	5	3
Italy	7	7	7	6
Japan	5	6	6	7
United Kingdom	2	2	2	2
United States	1	1	1	1

The analysis confirms the robustness of the top and bottom of the ranking. The United States and the United Kingdom consistently hold the first and second positions, respectively, across all scenarios, indicating their performance is strong regardless of policy priorities. Similarly, Italy and Japan generally remain in the lower tier. In contrast, the middle-tier rankings exhibit more sensitivity. Canada's rank, for example, improves from 5th to 3rd in both the 'Economy First' and 'Social First' scenarios, while Germany's rank declines from 3rd to 5th in the 'Social First' scenario, highlighting the competitive nature of this group.

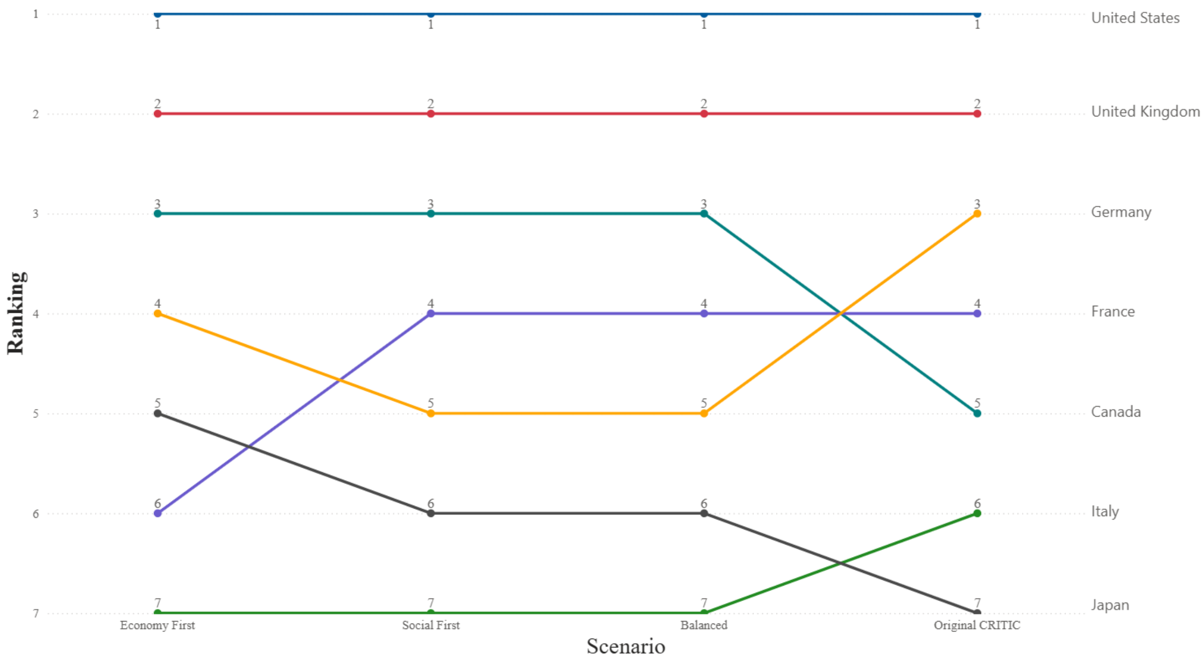

Figure 3. Robustness Check of G7 Rankings via Scenario Analysis

4. Discussion

This study's findings offer a multi-layered and nuanced perspective on the artificial intelligence landscape within the G7. While the final ranking provides a clear performance hierarchy, the true value of this analysis lies in deconstructing the underlying factors that drive these results. The discussion interprets the overall ranking, identifies divergent national strategic models, highlights the significance of the objective criteria weights, and outlines key policy implications.

4.1. Interpretation of the Overall Ranking: A Story of Stark Contrasts

The United States' position as the definitive leader (Score: 0.6648) is not surprising [2], but the data reveals its leadership is a story of stark contrasts. Its top rank is overwhelmingly driven by its unparalleled performance in the Economic Impact (Rank: 1) and Tech Readiness (Rank: 1) pillars. The sheer scale of its private investment (C1) and the volume of its AI publications (C12) create a significant gap with other nations. However, this growth-centric dominance comes at a demonstrable cost: the US ranks last in Policy & Governance (Rank: 7) and exhibits the highest income inequality (C6), a non-benefit criterion that received a high weight from the CRITIC analysis. This profile suggests a model of aggressive, market-driven innovation that may be outpacing its own regulatory and social equity frameworks.[69]

In second place, the United Kingdom (Score: 0.4514) emerges not merely as a runner-up, but as a proponent of a more balanced model. It is the only nation to secure a top-two position in both a growth-oriented pillar (Economic Impact, Rank: 2) and a society-oriented pillar (Social Impact, Rank: 1). This indicates a more integrated approach, suggesting that economic success in AI does not have to be mutually exclusive with strong social outcomes. The close grouping of Germany, France, and Canada behind the UK points to a competitive middle tier where national advantages are less pronounced. At the bottom, Italy and Japan face systemic challenges, ranking consistently low across most pillars, which indicates a need for a broad, foundational strategic overhaul rather than incremental improvements.

4.2. Divergent National Strategies: The 'Growth-First' vs. 'Governance-First' Models

The pillar-level analysis compellingly reveals that G7 nations are not on the same path; instead, they are pursuing fundamentally different strategic models for AI development.

The US "Growth-First" Model: This model prioritizes rapid innovation, commercialization, and technological supremacy. It fosters a dynamic ecosystem for investment and research but treats governance as a reactive, rather than proactive, element. The low rank in Policy & Governance is not an accident but a reflection of a culture that often favors permissionless innovation [70].

The European "Governance-First" Model: Embodied by Italy (Rank 1 in Policy), France (Rank 2), and Germany (Rank 3), this model prioritizes establishing robust ethical and legal frameworks, epitomized by regulations like the GDPR and the upcoming AI Act [71]. However, these nations have so far struggled to translate this regulatory leadership into commensurate economic or technological dynamism. Their challenge is to ensure that strong governance acts as a catalyst for "Trustworthy AI," rather than a brake on innovation [72].

The UK and Canada as "Balanced" Contenders: The UK, as discussed, balances economic and social aspects effectively. Canada presents another interesting hybrid, with strong Tech Readiness (Rank: 2) and respectable Economic Impact (Rank: 3), but, like the US, it lags in Policy & Governance (Rank: 6). This positions them as nations attempting to bridge the gap between the US and European models [73].

4.3. The Significance of Objective Weights: A Paradigm Shift in Assessment

A crucial insight from this study comes from the CRITIC method's objective weighting. The fact that the Data Protection Privacy Law Index (C9) and the Gini Coefficient (C6) emerged as the two most influential criteria is highly significant. This finding implies that, from a statistical standpoint, the greatest differentiating factors among the G7's AI ecosystems are not solely related to investment or publications, but are deeply rooted in regulatory maturity and social equity. It suggests that as AI technology becomes more widespread, the "hard problems" of governance and managing social impact become the primary arenas of national differentiation [74]. This elevates the discourse from a simple technological race to a more complex challenge of sustainable and equitable integration.

4.4. Policy Implications

The results offer several actionable insights for policymakers:

1. For Technologically-Leading Nations (e.g., the US): Sustainable leadership in the AI era will require more than just economic and technological might. A failure to proactively develop robust governance frameworks and address social inequality could undermine public trust and create long-term instability, ultimately hindering innovation [75].
2. For Governance-Leading Nations (e.g., the EU): The priority must be to create agile pathways to translate strong regulatory principles into a competitive advantage. This involves fostering sandboxes for innovation, streamlining funding for AI startups, and promoting the "Trustworthy AI" brand as a unique value proposition in the global market [76].
3. The Importance of a Balanced Scorecard: This research underscores the peril of policy "blind spots." Governments must adopt a holistic, multi-dimensional view of their AI ecosystem, similar to the four-pillar framework used here. Over-indexing on economic metrics while ignoring social readiness or governance can lead to a brittle and unsustainable AI strategy [77]. The scenario analysis confirmed the robustness of the top performers, indicating that a balanced, all-around strength is the most resilient strategy.

5. Conclusion

This study set out to develop and apply a comprehensive, objective framework for assessing the multifaceted artificial intelligence landscape across the G7 nations. By integrating the CRITIC method for objective weight determination with the TOPSIS method for performance ranking, this research successfully moves beyond single-metric evaluations to provide a holistic and data-driven comparative analysis.

5.1. Principal Findings

The research yields several key findings. First, the final ranking identifies the United States as the unequivocal leader, a position overwhelmingly driven by its profound strengths in the Economic Impact and Tech Readiness pillars. Following is the United Kingdom, which distinguishes itself with a more balanced strategic profile, achieving high ranks in both economic and social dimensions. Second, the analysis reveals a clear strategic divergence between G7 members, most notably contrasting the "growth-first" model of the US with the "governance-first" approach of European nations like Italy, France, and Germany. This highlights a fundamental trade-off between rapid innovation and the establishment of robust regulatory safeguards.

Perhaps the most significant contribution is the insight derived from the objective weighting process: the Data Protection Privacy Law Index and the Gini Coefficient emerged as the most statistically influential criteria. This finding compellingly suggests that in the current stage of AI development among advanced economies, the primary differentiators are shifting from purely technological and economic prowess to the more complex challenges of mature governance and social equity. The robustness of these core findings was further validated through a scenario analysis, which confirmed the stability of the top and bottom rankings across varying policy priorities.

5.2. Limitations

Despite the robustness of the methodology, this study has several limitations that should be acknowledged. First, the analysis is based on a static snapshot of data from a specific time frame. The AI field evolves at an exceptional pace, and a country's performance could shift rapidly due to new policy initiatives or technological breakthroughs [78]. Second, while the quantitative approach ensures objectivity, it cannot capture qualitative nuances, such as the practical effectiveness of a national AI strategy beyond its stated maturity, or the cultural and societal acceptance of AI technologies. Third, the study's scope is limited to the G7 nations. A complete global picture of the AI geopolitical landscape would necessitate the inclusion of other major players, most notably China [79].

5.3. Future Research

The findings and limitations of this study open up several promising avenues for future research. A logical next step would be to conduct a longitudinal analysis, applying this framework to data across multiple years to track national trajectories and measure the real-world impact of policy interventions. Furthermore, an interesting direction would be to develop a hybrid MCDM model that integrates qualitative data gathered from expert surveys (e.g., using the Delphi or AHP methods) to complement the objective quantitative findings presented here [80] [81]. Finally, expanding the framework to include a broader and more diverse set of countries would provide invaluable insights into the global dynamics of AI competition and cooperation, contributing to a more comprehensive understanding of the worldwide AI ecosystem.

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Competing interests

The authors have no relevant financial or non-financial interests to disclose.

Author contributions

All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by all authors. The first draft of the manuscript was written by Nhat-Luong Nhieu and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Data availability

All data generated or analyzed during this study are included in this published article.

Ethical approval

This article does not contain any studies with human participants performed by any of the authors.

Informed consent

This article does not contain any studies with human participants performed by any of the authors.

Appendix

Table 14. Initial Decision Matrix

	Code	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13
Canada	CA	2.89	19.69	0.731	13.5	21.86	0.2985	4	6	3	78.18	235.44	1.61	71.0
France	FR	2.62	8.40	-0.077	16.4	24.21	0.3066	4	14	5	79.35	307.96	1.56	35.0
Germany	DE	1.97	9.38	0.824	18.4	24.21	0.3243	3	13	5	76.90	98.33	2.80	40.9
Italy	IT	0.86	3.52	0.369	15.2	21.03	0.3524	2	15	5	71.22	96.32	1.68	8.2
Japan	JP	0.93	2.13	0.656	15.1	21.53	0.3234	3	14	4	75.75	214.9	2.79	16.0
United Kingdom	GB	4.52	21.96	0.809	11.7	26.65	0.3254	5	12	4	78.88	143.83	2.65	60.0
United States	US	109.08	24.45	1.349	10.2	23.71	0.4000	5	18	1	87.03	285.59	12.49	64.0

Table 15. CRITIC Normalized Matrix

	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13
Canada	0.0187581	0.7867384	0.5666199	0.5975610	0.1476868	1	0.6666667	0	0.5000000	0.4402277	0.6573427	0.0045746	1
France	0.0162632	0.2809140	0	0.2439024	0.5658363	0.9201970	0.6666667	0.6666667	1	0.5142315	1	0	0.4267516
Germany	0.0102569	0.3248208	0.6318373	0	0.5658363	0.7458128	0.3333333	0.5833333	1	0.3592663	0.0094973	0.1134492	0.5207006
Italy	0	0.0622760	0.3127630	0.3902439	0	0.4689655	0	0.7500000	1	0	0	0.0109790	0
Japan	0.0006468	0	0.5140252	0.4024390	0.0889680	0.7546798	0.3333333	0.6666667	0.7500000	0.2865275	0.5602911	0.1125343	0.1242038
UK*	0.0338200	0.8884409	0.6213184	0.8170732	1	0.7349754	1	0.5000000	0.7500000	0.4845035	0.2244850	0.0997255	0.8248408
US**	1	1	1	1	0.4768683	0	1	1	0	1	0.8943016	1	0.8885350

**United Kingdom*

***United States*

Table 16. CRITIC Correlation Matrix

Criteria	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13
C1	1	0.58761	0.69057	0.66940	0.11395	-0.85966	0.53448	0.57159	-0.86599	0.83172	0.45458	0.98926	0.42138
C2	0.58761	1	0.63773	0.76739	0.54066	-0.29046	0.87851	-0.16324	-0.73662	0.76510	0.29736	0.55437	0.94180
C3	0.69057	0.63773	1	0.59932	0.15858	-0.61231	0.43388	0.15203	-0.77700	0.57093	-0.05680	0.74957	0.54580
C4	0.66940	0.76739	0.59932	1	0.21324	-0.53976	0.71211	0.12440	-0.82531	0.61478	0.34824	0.64146	0.56824
C5	0.11395	0.54066	0.15858	0.21324	1	0.01804	0.69686	0.07737	-0.01301	0.46008	0.01061	0.13996	0.49811
C6	-0.85966	-0.29046	-0.61231	-0.53976	0.01804	1	-0.17035	-0.78141	0.60847	-0.46799	-0.04632	-0.87241	-0.02523
C7	0.53448	0.87851	0.43388	0.71211	0.69686	-0.17035	1	-0.02914	-0.64318	0.85505	0.55301	0.51351	0.83524
C8	0.57159	-0.16324	0.15203	0.12440	0.07737	-0.78141	-0.02914	1	-0.15064	0.27031	0.08107	0.60635	-0.39707
C9	-0.86599	-0.73662	-0.77700	-0.82531	-0.01301	0.60847	-0.64318	-0.15064	1	-0.80334	-0.51749	-0.84973	-0.64301
C10	0.83172	0.76510	0.57093	0.61478	0.46008	-0.46799	0.85505	0.27031	-0.80334	1	0.67324	0.81727	0.72094
C11	0.45458	0.29736	-0.05680	0.34824	0.01061	-0.04632	0.55301	0.08107	-0.51749	0.67324	1	0.38898	0.34912
C12	0.98926	0.55437	0.74957	0.64146	0.13996	-0.87241	0.51351	0.60635	-0.84973	0.81727	0.38898	1	0.38781
C13	0.42138	0.94180	0.54580	0.56824	0.49811	-0.02523	0.83524	-0.39707	-0.64301	0.72094	0.34912	0.38781	1

Table 17. TOPSIS Normalization Matrix

	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13
Canada	0.02645	0.48573	0.35417	0.34977	0.35330	0.33735	0.39223	0.16705	0.27735	0.37733	0.41774	0.11789	0.56204
France	0.02398	0.20722	-0.03731	0.42490	0.39128	0.34650	0.39223	0.38979	0.46225	0.38298	0.54641	0.11423	0.27706
Germany	0.01803	0.23140	0.39923	0.47672	0.39128	0.36651	0.29417	0.36195	0.46225	0.37115	0.17447	0.20503	0.32377
Italy	0.00787	0.08684	0.17878	0.39381	0.33989	0.39827	0.19612	0.41763	0.46225	0.34374	0.17090	0.12302	0.06491
Japan	0.00851	0.05255	0.31783	0.39122	0.34797	0.36549	0.29417	0.38979	0.36980	0.36560	0.38130	0.20430	0.12666
United Kingdom	0.04137	0.54173	0.39196	0.30313	0.43072	0.36775	0.49029	0.33411	0.36980	0.38071	0.25520	0.19405	0.47496
United States	0.99828	0.60316	0.65359	0.26427	0.38320	0.45206	0.49029	0.50116	0.09245	0.42004	0.50672	0.91460	0.50663

Table 18. TOPSIS Weighted Matrix

	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13
Canada	0.001745	0.032195	0.021843	0.021731	0.025369	0.040885	0.022345	0.013376	0.041582	0.017111	0.036244	0.007577	0.037996
France	0.001582	0.013735	-0.002301	0.026399	0.028097	0.041995	0.022345	0.031210	0.069304	0.017367	0.047408	0.007342	0.018730
Germany	0.001189	0.015337	0.024622	0.029618	0.028097	0.044419	0.016759	0.028981	0.069304	0.016831	0.015137	0.013178	0.021888
Italy	0.000519	0.005756	0.011026	0.024467	0.024406	0.048268	0.011173	0.033439	0.069304	0.015587	0.014828	0.007907	0.004388
Japan	0.000561	0.003483	0.019602	0.024306	0.024986	0.044296	0.016759	0.031210	0.055443	0.016579	0.033082	0.013131	0.008562
United Kingdom	0.002729	0.035907	0.024174	0.018833	0.030928	0.044570	0.027932	0.026752	0.055443	0.017264	0.022141	0.012472	0.032109
United States	0.065849	0.039978	0.040310	0.016419	0.027516	0.054788	0.027932	0.040127	0.013861	0.019048	0.043964	0.058784	0.034250

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